

Trip Generation Table - 12th Edition

LUC ¹	Land Use	Unit	ADT	AM Peak Hour			PM Peak Hour		
				Enter	Exit	Total	Enter	Exit	Total
Total Site Trips ²									
110	Light Industrial	150,000 sf	540	62	10	72	18	56	74

Notes:

1. Land Use Code and trip generation rates are determined based on ITE Trip Generation, 12th Edition
2. Total site trips are determined based on the ITE average rates

ITE 12th Edition Average Rates - LUC 110 Light Industrial (p

Period	Rate	% Enter
Weekday (ADT)	3.60	50%
AM Peak Hour	0.48	86%
PM Peak Hour	0.49	24%

Source: ITE Trip Generation Manual, 12th Edition. Blue cells are inputs; the TripGen table

City of Durham TIA Threshold Check (UDO Section 3.3)

Peak hour trip threshold	150	
Max site peak hour trips	74	
Result	TIA Not Require	

er 1,000 sf GFA)

% Exit
50%
14%
76%

calculates from these rates.

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Draft Site Trip Assignment - Turn Lane Evaluation (pre-count)

Directional Distribution (blue inputs; draft, AADT-weighted)

To/from south via Page Rd		35%
To/from north via Page Rd		30%
To/from southeast via Airport Rd / Pleasant Grove Church Rd (Aviation Pkwy / I-40)		35%
Sum		100%

Peak Hour Site Trips (from TripGen)

			AM In	AM Out	PM In	PM Out
Control totals			62	10	18	56

Driveway Movement Assignment

Access / Movement	Serves	Share	AM In	AM Out	PM In	PM Out
Page Rd Dwy (auto, RIRO): NB right-in	From south	35%	22	-	6	-
Page Rd Dwy (auto, RIRO): right-out to north	To north	30%	-	3	-	17
Airport Rd Dwy (auto): EB right-in	From north (via int.)	30%	19	-	5	-
Airport Rd Dwy (auto): WB left-in	From SE (via PGC Rd)	35%	21	-	7	-
Airport Rd Dwy (auto): WB left-out (to south via int.)	To south	35%	-	4	-	20
Airport Rd Dwy (auto): EB right-out	To SE (via PGC Rd)	35%	-	3	-	19
Truck driveways (Page Rd S dwy / Airport Rd E dwy)	Trucks, off-peak	-	0	0	0	0
Total			62	10	18	56

Notes:

1. Distribution is a draft engineering input pending the TMC; proportional to background volumes (Page Rd S 8,800 / Page Rd N 7,100 / Airport Rd-PGC Rd corridor 9,10
2. Peak hour trips assigned to the two auto driveways; truck driveways assumed negligible during peak hours.
3. Page Rd driveways assumed right-in/right-out only; Airport Rd driveways assumed full movement. Arrivals from the north and departures to the south route through

Turning Movement Count Entry - Page Rd (SR 1973) at Airport Rd (SR 1971)

Count date	TBD
Count year	2026

Approach	Movement	AM Peak	PM Peak
Page Rd NB (from south)	NBT (continue north)	0	0
	NBR (to Airport Rd)	0	0
Page Rd SB (from north)	SBL (to Airport Rd)	0	0
	SBT (continue south)	0	0
Airport Rd WB	WBL (to Page Rd south)	0	0
	WBR (to Page Rd north)	0	0

Blue cells are inputs. Enter peak hour turning movement volumes from the TMC; downstream sheets update automatica

Peak Hour Volume Development - Existing / No-Build / Build

Growth factor by approach: $(1 + \text{selected rate})^{(\text{Build year} - \text{Count year})}$

Page Rd south leg (NB approach)	1.061
Page Rd north leg (SB approach)	1.061
Airport Rd (WB approach)	1.061

Location	Movement	Exist AM	Exist PM	No-Build AM	No-Build PM	Site AM	Site PM	Build AM	Build PM
Page Rd @ Airport Rd	NBT	0	0	0	0	3	17	3	17
	NBR	0	0	0	0	0	0	0	0
	SBL	0	0	0	0	19	5	19	5
	SBT	0	0	0	0	0	0	0	0
	WBL	0	0	0	0	4	20	4	20
	WBR	0	0	0	0	0	0	0	0
Dwy P1 (Page Rd, RIRO)	NB right-in	0	0	0	0	22	6	22	6
	Right-out (to north)	0	0	0	0	3	17	3	17
Dwy A1 (Airport Rd)	EB right-in	0	0	0	0	19	5	19	5
	WB left-in	0	0	0	0	21	7	21	7
	WB left-out	0	0	0	0	4	20	4	20
	EB right-out	0	0	0	0	3	19	3	19

Warrant Analysis Inputs (Build volumes)

Warrant location	Type	Adv/Opp AM	Adv/Opp PM	Turn AM	Turn PM
Page Rd NB at Dwy P1 (right-in)	RT	22	6	22	6
Airport Rd EB at Dwy A1 (right-in)	RT	19	5	19	5
Page Rd NB at intersection (NBR)	RT	3	17	0	0
Airport Rd WB left-in at Dwy A1	LT	19	5	21	7
Intersection SBL (Page Rd SB)	LT	3	17	19	5
Intersection WBL (Airport Rd WB)	LT	19	5	4	20

Notes:

1. No-Build = Existing x growth factor for the movement approach. Build = No-Build + site trips (Assign sheet).
2. Site trips at the intersection: NBT gains P1 right-out (to north); SBL gains arrivals from the north routing to Dwy A1; WBL gains departures to the south. Other movements unchanged.
3. Advancing volumes are directional (through + turns approaching the location). Opposing volume for left turns = oncoming directional volume at the location.
4. Dwy P1 advancing (row 26) = No-Build NB leg + site right-in (site right-in turns off before the intersection).

NCDOT AADT Data & Growth Projections - Page Road (SR 1973)

NCDOT AADT History (vpd)

Location (TMS Station)	2013	2015	2017	2019	2021	2023
Page Rd south of Airport Rd (0320000269)	6,700	7,700	9,900	-	10,000	8,800
Page Rd north of Airport Rd (0320000260)	8,000	5,500	6,800	8,200	6,600	7,100
Airport Rd / Pleasant Grove Church Rd (0920000468)	7,300	7,800	-	-	-	9,100

Growth Rates & Build Year Projection

Location	2023 AADT	CAGR '13-'23	CAGR '17-'23	Selected Rate	Build AADT	Build AADT (rounded)
Page Rd south of Airport Rd	8,800	2.8%	-1.9%	3.0%	10,202	10,200
Page Rd north of Airport Rd	7,100	-1.2%	0.7%	3.0%	8,231	8,200
Airport Rd / PGC Rd corridor (SR 1789)	9,100	2.2%	-	3.0%	10,549	10,500
Assumed opening year		2027				
Build (analysis) year		2028				

Per City of Durham TIA Guidelines: analysis examines conditions one year after project completion

Airport Rd / Pleasant Grove Church Rd Corridor Estimate

SR 1789 segment AADT near SE end of corridor (2023)	9,100
Difference method at Page Rd intersection (minimum)	1,700
Projected to Build year at selected rate	10,500

Notes:

- 1. Source: NCDOT AADT stations 0320000269 (Page Rd south of Airport Rd), 0320000260 (Page Rd north of Airport Rd), and 0920000468 (SR 1789 / Pleasant Grove Church Rd, Wake Co.),
- 2. Selected growth rate is an input (blue cells).
- 3. Corridor estimate uses the SR 1789 segment AADT (9,100, 2023, source "Supplemental") near the SE end of the corridor; limited development between the segment and the site. Difference method at the

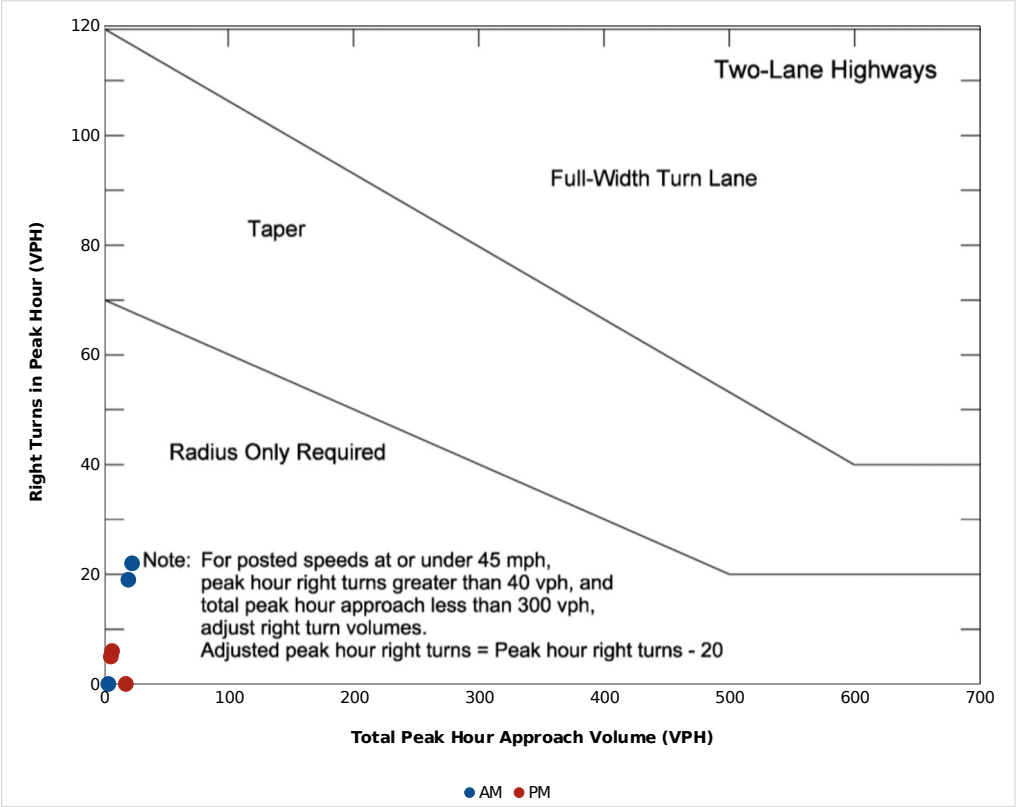
Development Summary

Project	SCP Page Road Industrial Development
Site Address	3905 Page Rd, Durham, NC
Location	Page Road (SR 1973) at Airport Road (SR 1971)
Client	Strategic Capital Partners, LLC
VHB Project #	R205114.000
Concept Plan Building	132,080 sf industrial warehouse (520' x 260'), ±133 auto stalls, 24 dock positions
Concept Plan Source	ARCO Design/Build, Scheme 01, dated 03/23/2026
Analyzed Building Size	Up to 150,000 sf GFA (conservative buffer for changes to the building)
Proposed Access	Two driveways on Page Road and two driveways on Airport Road
ITE Land Use Code	LUC 110 - Light Industrial (ITE Trip Generation Manual, 12th Edition)

NCDOT Right Turn Lane Warrant - Two-Lane Highways (Roadway Design Manual, May 2026, Figure 8-9)

Location	Period	Posted Speed (mph)	Approach Volume (vph)	Right Turns (vph)	Adjusted RT ¹	Full-Lane Line	Taper Line	Determination
Page Rd NB at Dwy P1	AM	45	22	22	22	117	68	Radius only
Airport Rd EB at Dwy A1	AM	45	19	19	19	117	68	Radius only
Page Rd NB at intersection (NBR)	AM	45	3	0	0	120	70	Radius only
Page Rd NB at Dwy P1	PM	45	6	6	6	119	69	Radius only
Airport Rd EB at Dwy A1	PM	45	5	5	5	119	70	Radius only
Page Rd NB at intersection (NBR)	PM	45	17	0	0	118	68	Radius only

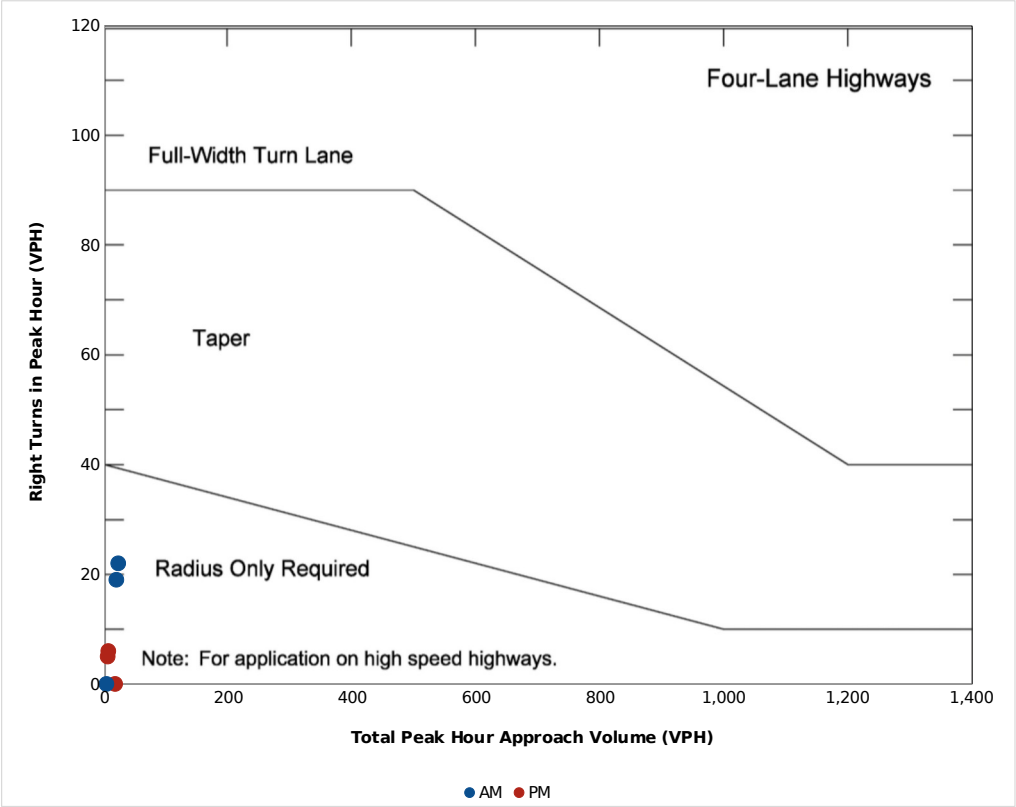
- Notes:
- Adjusted RT = RT - 20 when posted speed <= 45 mph, RT > 40 vph, and approach volume < 300 vph (chart note). Blue speed cells are inputs.
 - Volumes are Build peak hour values from the Volumes sheet. Determination compares the adjusted right turns to the digitized warrant boundary lines.
 - Chart background is the NCDOT figure (source: NCHRP 279 Figure 4-23). Two-lane chart applies to the existing two-lane sections of Page Rd and Airport Rd.



NCDOT Right Turn Lane Warrant - Four-Lane Highways (Roadway Design Manual, May 2026, Figure 8-9)

Location	Period	Posted Speed (mph)	Approach Volume (vph)	Right Turns (vph)	Adjusted RT ¹	Full-Lane Line	Taper Line	Determination
Page Rd NB at Dwy P1	AM	45	22	22	22	90	39	Radius only
Airport Rd EB at Dwy A1	AM	45	19	19	19	90	39	Radius only
Page Rd NB at intersection (NBR)	AM	45	3	0	0	90	40	Radius only
Page Rd NB at Dwy P1	PM	45	6	6	6	90	40	Radius only
Airport Rd EB at Dwy A1	PM	45	5	5	5	90	40	Radius only
Page Rd NB at intersection (NBR)	PM	45	17	0	0	90	39	Radius only

- Notes:
- 1. Adjusted RT = RT - 20 when posted speed <= 45 mph, RT > 40 vph, and approach volume < 300 vph (chart note). Blue speed cells are inputs.
 - 2. Volumes are Build peak hour values from the Volumes sheet. Determination compares the adjusted right turns to the digitized warrant boundary lines.
 - 3. Four-lane chart provided for a future widened condition (proposed thoroughfare along Page Rd); not applicable to the existing two-lane sections.



NCDOT Warrant for Left (and Right) Turn Lanes - Storage Nomograph (Policy on Street and Driveway Access, July 2000)

Location	Period	Left Turns (vph)	Opposing Volume (vph)	Storage S (ft, chart read)	Warrant read
Airport Rd WB left-in at Dwy A1	AM	21	19	-	Below S=50 - no storage indicated
Intersection SBL (Page Rd SB)	AM	19	3	-	Below S=50 - no storage indicated
Intersection WBL (Airport Rd WB)	AM	4	19	-	Below S=50 - no storage indicated
Airport Rd WB left-in at Dwy A1	PM	7	5	-	Below S=50 - no storage indicated
Intersection SBL (Page Rd SB)	PM	5	17	-	Below S=50 - no storage indicated
Intersection WBL (Airport Rd WB)	PM	20	5	-	Below S=50 - no storage indicated

Notes:

- 1. Plot each point (left turns vs. opposing volume) on the nomograph. A point below/left of the S=50 curve indicates no left turn storage is required; otherwise read the required
- 2. Right turn storage may also be read from this chart (opposing volume of 100 vph minimum per the figure note).
- 3. Volumes are Build peak hour values from the Volumes sheet. Where adjacent signalization provides gaps, a storage reduction may be considered case-by-case (figure note).

